

Answer key Exercise ISU-19: Using Electronic Leak Detectors

Source: Solution and Answer Guide, Silberstein/Obrzut, The Complete HVACR Manual 2E, Section 13: Installation and Start-Up (ISU), and the ISU-19 lab sheet.

Context

Objectives: Upon completion of this exercise, you should be able to use an electronic leak detector to find refrigerant leaks.

Introduction: You will check an air-conditioning system with gauges to determine that there is refrigerant pressure. You will then check all field and factory connections with an electronic leak detector to determine if there are refrigerant leaks.

Takeaway: Electronic leak detectors can be highly sensitive, detecting leaks as small as about 1/4 oz a year. They are capable of picking CFC, HCFC, and HFC refrigerants, depending on the model. When using a leak detector, check all field and factory connections on the system.

Exercise ISU-19

ISU-19 Summary statement

Describe how the electronic leak detector operates. What are some of its good and bad points?

Model answer: A small pump in the detector provides suction that pulls air, and any refrigerant from a leak, to the sensor. A filter under the tip keeps airborne particles out of the sensing element. When refrigerant is sensed, the alarm goes off. It runs on batteries or 115 V AC. Good points: it is highly sensitive, detecting leaks as small as about 1/4 oz a year; depending on the model it picks up CFC, HCFC and HFC refrigerants; it works in a light or a dark place; most models have a sensitivity adjustment to pinpoint the source of a large leak. Bad points: wind affects leak detection, so the area must be shielded; it cannot be used to locate natural gas leaks because ignition may occur; with a large leak the alarm can go off before the probe reaches the source unless the sensitivity is turned down.

Not in the official answer key. Composed only from the Takeaway on the lab sheet and model answers 1 to 8.

Question ISU-19.1

Can natural gas leaks be located with an electronic leak detector?

Model answer: No, ignition may occur.

Question ISU-19.2

How can the refrigerant be moved from the point of the leak to the detector sensor?

Model answer: With a small pump to provide a suction in the electronic leak detector.

Question ISU-19.3

What can be done to keep wind from affecting leak detection?

Model answer: Shield the area where you are attempting to find a leak.

Question ISU-19.4

What is the power source for an electronic leak detector?

Model answer: Batteries or 115 V AC.

Question ISU-19.5

How can soap bubbles be used to further pinpoint the leak when it is detected?

Model answer: Soap bubbles will form a bubble at the point of the leak.

Question ISU-19.6

What is the purpose of the filter under the tip of the sensor on the detector?

Model answer: To prevent airborne particles from being sucked into the sensing element.

Question ISU-19.7

Does the electronic leak detector work best in a light or dark place?

Model answer: Either.

Question ISU-19.8

How can an electronic leak detector be used to pinpoint the source of a large leak?

Model answer: Most models have a sensitivity adjustment that can be used when there is a large leak. The sensitivity can be turned down to prevent the alarm from going off before it reaches the source of the leak.